FAV Group of Companies



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目 录
Catalogs

1、公司简介 ----------2
Company Profile

2、JC738-SF9 型伺服控制系统九单元双浆槽浆纱机---------- 4
JC739-SF9 Servo control system nine-units drive high-speed
high efficiency intelligent double sizing-boxs machine

3、各组合单元技术配置 ----------7
Technical configuration of each combined unit

4、JC739SF-D6 伺服控制系统六单元单浆槽浆纱机 ---------- 11
JC739SF-D6 Servo control system six-unit drive high-speed
high efficiency intelligent one sizing-box machine

5、JC739-S9 型变频控制系统七单元、九单元双浆槽浆纱机 ---------- 12
JC739-S9 Inverter control system nine-unit electromechanical
integrated double sizing-boxs sizing machine

6、JC739-D6 型变频控制系统六单元单浆槽浆纱机 ---------- 13
JC739-D6 Six-unit of the frequency conversion control system
technical configuration of ones sizing-box size machine

公 司 简 介

高密市家诚机械科技有限公司（原山东鸿基纺机）始于 1995 年，是专业从事浆纱设备研发与生产的专业型企业。为客户提供伺服控制系统、变频控制系统等多单元控制的各系列单、双浆槽浆纱机。

公司建立以来，始终坚持科技发展的战略，同行业重点科研机构建立了良好的合作关系。通过自动化和智能化的升级，实现了浆纱设备的高速、高效、节能、低耗，同时降低了对操作★员的技术要求。

公司一直奉行“科技创新，优质服务”的经营理念，以高科技、高可靠性和最优质的服务为国内外纺织企业提供精良的技术装备。

高密市家诚机械科技有限公司期待与您携手合作，共创辉煌！



Company Profile

Gaomi Jiacheng Textile Machinery Technology Co., Ltd (original Shandong Hongji Textile Machinery) started in 1995, is a professional enterprise specialising in the research, development and production of textile sizing equipment. To provide customers with servo control system, frequency-control system and other multi-unit control of the series of single and double sizing machine.

Since its establishment, the company has always adhered to the strategy of science and technology development, and established a good cooperative relationship with key scientific research institutions in the industry. Through the upgrade of automation and intelligence, the high speed, high efficiency, energy saving and low consumption of the sizing equipment are realized, and the technical requirements for the operators are reduced.

The company has been pursuing the "scientific and technological innovation, quality service" business philosophy, with high-tech, high reliability and the best quality service to domestic and foreign textile enterprises to provide excellent technical equipment.

Gaomi Jiacheng Machinery Technology Co., Ltd . is looking forward to working with you to create brilliant!

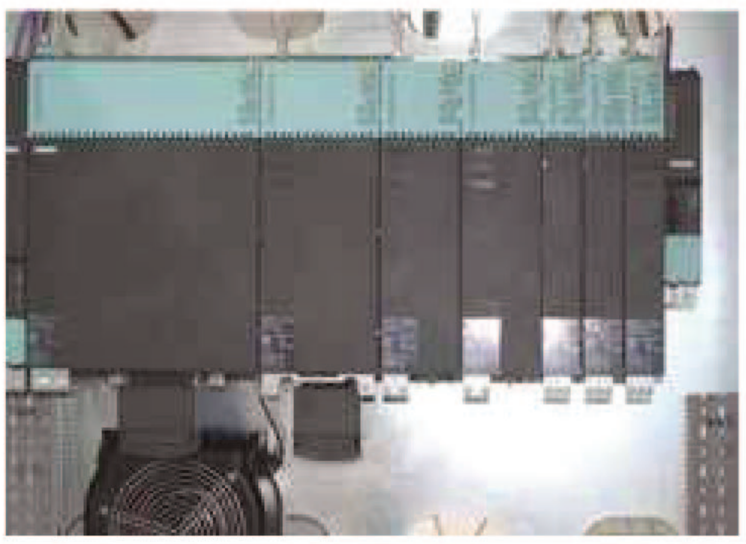


JC739-SF9 型伺服控制系统九单元 传动高速、高效、智能型双浆槽浆纱机

JC739-SF9 Servo control system
Nine-units drive high-speed high efficiency
intelligent double sizing-boxs machine

1 产品的主要技术参数 Main technical parameters of the product

项目 project	单位 Units	JC739-SF9
车头有效幅宽 The front is effective and wide	mm	2000/2400/2800/3600/4200
适应品种范围 Adapt to variety range		6S-200S 短纤维经纱 Short fiber through yarn
车速 speed	m/min	1-200
蜗牛速度 Snail speed	m/min	0.5
织轴卷径 Max weaer’ beam diameter	mm	800/1000/1250
最大卷绕张力 Max. around tension	KN	7.5
最大压力 max. pressure	KN	55
经轴幅宽 Warper’sbeam diameter	mm	1800/2000/2400/2800
烘筒数量 The number of drying cylinder		14/20
容纳经轴数量 Number of warper’ s beams		24（可据要求）
经轴最大直径 Max. diameter through warper’ s beams	mm	800/1000/1250
重量（约） Weight (approx)	T	35





整机主要技术特点

1、九单元伺服同步传动调速系统

采用目前技术最先进、可靠性指标最好的传动技术——多单元同步伺服传动、调速、张力检测与数字控制技术。双浆槽全烘筒式浆纱机共设九个传动单元。

A、采用多单元伺服同步传动调速系统，即使各张力区的纱线张力有微小变化，高灵敏度的张力传感器和同步伺服传动调速系统都可在线检测并反馈，并准确地进行实时调节。

B、很宽范围内的张力稳定性和可调节性，对品种的适应范围进一步提高。无论对于需较大张力控制的粗纱品种，还是对张力敏感的细纱品种，均可取得良好的质量。

C、采用多单元同步伺服传动，设备的机械结构大为简化，平时所需的维护保养工作变得更少也更方便。

机械传动的振动和噪声明显减少，设备运转更加平稳。尤为重要的是设备传动系统的可靠性指标大幅度提高。

D、模块化设计多单元传动，使设备的组合式水准有了提高。

满足不同的使用要求，在设备结构功能上为用户提供了更多的可选择性，也为用户今后设备升级提供了方便。

2、工业用计算机监控系统

⑴ 采用高性能工业用计算机集中监控系统，各项工艺参数均可预先设定，对于影响浆纱质量的各种干扰源，计算机通过在线检测、判断或运算后发出控制指令，把设备的各项工艺参数或运行状态调整到最佳状态。

⑵ 各种工艺参数具有高度的重现性，经过实际生产验证和优选的上浆工艺可贮存于计算机系统，当今后再生产同类品种时，可随时调用并可获得同等的质量指标。

⑶ 采用的大屏幕 TFT 彩色终端，具有设备运行状态数据与图形显示，工艺参数的预先设定与运行中的修改，故障的诊断与报警等一系列的工业现场的人机对话功能，使得对设备的操作和控制更为方便与主

Main technical features of the whole machine

1. Nine-units servo synchronous drive speed control system

Using the most advanced technology, the best reliability index of the transmission technology —— multi-unit synchronous servo transmission, speed regulation, tension detection and digital control technology. The machine is equipped with nine transmission units.

A. The multi-unit servo synchronous transmission speed regulation system is adopted. Even if the yarn tension in each tension area changes slightly, the high sensitive tension sensor and synchronous servo transmission speed regulation system can be detected online and feedback, and accurately adjusted in real time.

B, Tension stability and adjustable in a wide range, further improve the adaptation range of the varieties. Good quality can be obtained for both the roving yarn varieties that need large tension control and the fine spinning yarn varieties that are sensitive to tension.

C, the use of multi-unit synchronous servo transmission, the mechanical structure of the equipment is greatly simplified, and the usual required maintenance work becomes less and more convenient.

The vibration and noise of the mechanical transmission are significantly reduced, and the equipment is running more smoothly. Most importantly, the reliability index of the equipment transmission system is greatly improved.

D, the modular design of multi-unit transmission, so that the combination of the equipment level has been improved.

It meets different use requirements, provides users with more options in the device structure and function, and also provides convenience for users

动。

⑷ 采用计算机集中监控系统，设备可实现的自动化功能有：

a. 浆液温度在线检测与自控；

b. 压浆力在线检测与自控；

c. 浆液液位在线检测与自控；

d. 烘燥温度在线检测与自控；

e. 回潮率在线检测与自控；

f. 伸长率在线检测与自控；

g. 经纱退绕张力的在线检测与自控；

h. 喂入伸长的在线检测与自控；

i. 湿区伸长的在线检测与自控；

j. 干区张力的在线检测与自控；

k. 卷绕张力的在线检测与自控；

3、独立传动的侧压式高压浆槽

上浆机构的特点：

⑴ 采用侧压式高压上浆系统可使浆纱获得满意的渗透与被覆效果。由于在浸压数量上具有较大的选择余地，提高了对不同纱线品种的工艺适应性。

⑵ 良好的浆液流动性措施和高精度的浆液温度自控，为提高上浆质量提供了浆液条件。

⑶ 可达 5500Kgf 的高压浆力指标，不仅使设备节能、高效，而且提高了浆纱质量。

⑷ 采用的与上浆辊同挠度方向的橄榄芯结构压浆辊提高了上浆率的横向一致性。

⑸ 由计算机控制的压浆力自控系统，其实际压浆力可以按照任意的“速度——压浆力”曲线控制，提高了上浆率的纵向一致性。

⑹ 浆槽罩壳采用快卸式结构，使对浆槽的维护、保养更为方便。

4、独特的高精回潮率自控系统

（1） 开车一键升速

（2） 回潮率自动控制，自动升降速

（3） 多种纱线回潮预设模式

（4） 高性能回潮仪提供精确控制，车速自控，确保浆纱质量。

to upgrade their equipment in the future.

2, industrial computer monitoring system

(1) The centralized monitoring system of high-performance industrial computer is adopted, and all process parameters can be set in advance. For various interference sources affecting the quality of yarn, the computer issues control instructions after online detection, judgment or operation, and adjusts the process parameters or operation state of the equipment to the best state.

(2) Various process parameters have a high degree of reproducibility, after the actual production verification and preferred the pulping process can be stored in the computer system, now after the same varieties, can be called at any time and the same quality index can be obtained.

(3) The large screen TFT color terminal has a series of man-machine dialogue functions in the industrial field, such as equipment operation state data and graphic display, pre-setting and modification of process parameters in operation, fault diagnosis and alarm, which makes the operation and control of the equipment more convenient and active.

(4) Using the computer centralized monitoring system, the equipment can realize the automatic functions are:

A. Online detection and automatic control of slurry temperature;

B. Online detection and self-control of the grouting force;

C. Online detection and automatic control of serous level;

d. Online detection and self-control of drying temperature;

E. Online detection and self-control of moisture return rate;

f. Online detection and automatic control of elongation;

G. Online detection and self-control of the warp yarn unwinding tension;

H. Online detection and self-control of feeding and elongation;

I. Online detection and self-control of wet area elongation;

J. Online detection and self-control of dry area tension;

K. Online detection and self-control of winding tension;

3, the independent transmission of the side pressure type high-pressure slurry groove

Characteristics of the sizing mechanism:

(1) The side pressure high pressure grouting system can make the yarn achieve satisfactory penetration and coating effect. Due to the large choice for the number of dip pressure, the process adaptability for different yarn varieties is improved.

(2) Good slurry fluidity measures and high-precision slurry temperature control provide slurry conditions for the improvement of slurry quality

2 各组合单元技术配置
Technical configuration of each combined unit

（一）经 轴 架:

1、双层组合式，托轮式结构，每四只经轴为一组,每组之间留有人行踏板 ，便于操作；

2、采用经纱退绕张力气动控制系统，实现张力自动控制。

3、采用穿芯轴导辊，辊面跳动小，阻力小，运行平稳。

4，采用高强度尼龙及双轴承支撑，运行平稳，故障率低。

(I) Beam creel:

1. Double layer combination, wheeled structure, each four axis for a group, each group between the pedestrian pedal, easy to operate;

2. Use the warp unwinding strength motion control system to realize automatic tension control.

3. the use of the mandrel guide roller, roller surface beating is small, small resistance, smooth operation.

4. using high strength nylon and double bearing support, stable operation, low failure rate.



（二）、浆槽：

1、浆槽结构：

（1） 四辊式浆槽，浆槽机架为钢板结构，外罩不锈钢罩壳；

（2） 主浆槽容积 220L，辅助浆槽容积 200L；

（3） 主浆槽与预热浆槽联成一体，浆槽槽体为不锈钢双层结构，辅助浆槽靠烘房一侧；

（4） 槽底前方进浆，瀑布式溢流口回浆，浆液循环采用不锈钢齿轮浆泵，浆泵流量 20-50L/min，逆流式循环，粘度均匀，温度稳定。

（5） 蒸汽直接加热，配备温度自动控制系统；

（6） 采用引纱辊张力辊和压纱辊三辊式引纱装置，引纱辊表面包覆合成橡胶，压纱辊气动加压；

（7） 配置两根湿分绞，速度随车速变化和变化；

（8） 辅助浆槽带有浆液过滤。

2 上浆机构加压方式及压力指标：

(1) 四辊式双浸三压上浆形式，品种适应性强；

(2) 前后压浆辊均采用气动双侧同步加压方式，压浆力可随车速变化无级线性加压，前压浆辊为橄榄芯式高压压浆辊，使上浆率均匀稳定；压浆辊：外包进口合成橡胶，。

(3) 浸没辊采用气动双侧同步升降方式，浸没辊带侧压，最大压力为 3KN。

(4) 压浆力指标：

前压浆辊压力：0-50KN，压力随车速变化而呈线性变化；

后压浆辊压力：0-30KN，压力可无级设定；

3、浆槽传动：引纱辊和上浆辊各由一只独立伺服电机传动；

4、具有退绕张力自动控制系统。

(3) The main slurry groove is integrated with the preheated slurry tank. The groove body of the slurry groove is a stainless steel double-layer structure, and the auxiliary slurry groove is placed on the side of the oven;

(4) the front of the bottom of the groove slurry, waterfall overflow outlet grouting, slurry circulation using stainless steel gear slurry pump, slurry pump flow 20-50 L/min, counterflow cycle, uniform viscosity, stable temperature.

(5) Direct steam heating, equipped with automatic temperature control system;

(6) Using the three-roll lead device, the surface of the lead roll is coated with synthetic rubber, and the pressure roller is pneumatic pressurized;

(7) Equipped with two wet partiings, the speed changes and changes with the speed;

(8) Auxiliary slurry tank with slurry filtration.

2 Pulping mechanism, pressure mode and pressure index:

(1) Four-roller double immersion and three-pressure slurry form, strong variety adaptability;

(2) The front and rear grouting roller adopts pneumatic bilateral synchronous pressure mode, the grouting force can change with the speed of stepless linear pressure, the front grouting roller is olive core type high pressure grouting roller, make the sizing rate uniform and stable; the grouting roller: imported synthetic rubber.

(3) The immersed roller adopts the pneumatic synchronous lifting mode on both sides, and the immersed roller belt has the side pressure, and the maximum pressure is 3 KN.

(4). Pressure slurry force index:

Front pressure: 0-50 KN, the pressure changes linearly with the speed;

Back pressure slurry roller pressure: 0-30 KN, the pressure can be set stepless;

3, slurry groove drive: the yarn roller and

(II) Sizing box:

1. Structure of the size:

(1) four roll type slurry groove, slurry groove frame for steel plate structure, cover stainless steel shell;

(2) The main slurry tank volume is 220L, and the auxiliary slurry tank volume is 200L;

the slurry roller are driven by an independent servo motor;

4, with the automatic control system of unwinding tension.





（三）、烘　房：

1、烘房结构：　高架一列式结构，　采用直径 800mm 的不锈钢烘筒，共 20 支烘筒，其中 16 支烘筒表面喷涂聚四氟乙稀防粘层；

2、烘房传动：预烘烘筒和合并烘干烘筒分别由三只独立伺服电机传动；

3、穿纱路线：湿纱可分2 层或4 层进入烘房预烘；

4、温度控制：预烘烘房和合并烘干烘房共配备 5 套温度自动控制系统；

5、烘房排湿：烘房与浆槽的排湿汽罩为直角梯形防滴漏不锈钢汽罩，侧面用 3 只轴流风机强迫排湿；

6、烘房门薄板用不锈钢制作。

(III) Drying room:

Roof structure: elevated column structure, using stainless steel drying cylinder with a diameter of 800mm, a total of 20 drying cylinders, 16 drying surface spray polytetrafluoroethylene thin anti-adhesive layer;

Broom drive: pre-dr ying cylinder and combined drying cylinder are respectively driven by three independent servo motors;

Through yarn route: wet yarn can be divided into 2 layers or 4 layers into the oven pre-drying;

Temperature control: the pre-drying room and the combined drying room are equipped with 5 sets of automatic temperature control system;

Room exhaust humidity: the exhaust hood of the oven and the slurry tank is a right Angle trapezoidal anti-drip stainless steel hood, the side with 3 axial flow fan forced exhaust wet;

The baked door sheet is made of stainless steel.

（四）、车头：

1、传动： 织轴卷绕传动采用同步伺服电机独立单元恒张力卷绕系统，可实现锥形张力控制；

2、织轴加压、测长辊加压、拍合、上落轴等动作均由气动控制。

3、采用测长辊测长；

4、电子计长喷墨打印；

5、配备纱线单面上蜡及预融装置；

6、拖引辊为橡胶辊，由独立同步伺服电机传动；

7、采用双罗拉式侧面压纱装置， 最大压力3KN；

8、配备新型的全密封电动伸缩筘，易操作、防飞花；

9、分绞棒一批，按经轴数量配备；

10、织轴箱采用齿轮传动（公司发明专利）；

11、干分纱区使用张力辅助控制系统（公司发明专利）。

12、具有回潮自动控制及一键升速功能。



(Ⅳ) Headstock:

1, transmission: spindle winding transmission adopts synchronous servo motor independent unit constant tension winding system, which can realize conical tension control;

2, weaving axis pressure, measuring roller pressure, closing, falling shaft and other actions are controlled by pneumatic.

3, the length measuring roll measurement;

4, electronic meter long inkjet printing;

5. Equipped with yarn single-side wax and pre-melting device;

6. The towing roller is a rubber roller, driven by an independent synchronous servo motor;

7, using double roller type side yarn pressing device, the maximum pressure of 3 KN;

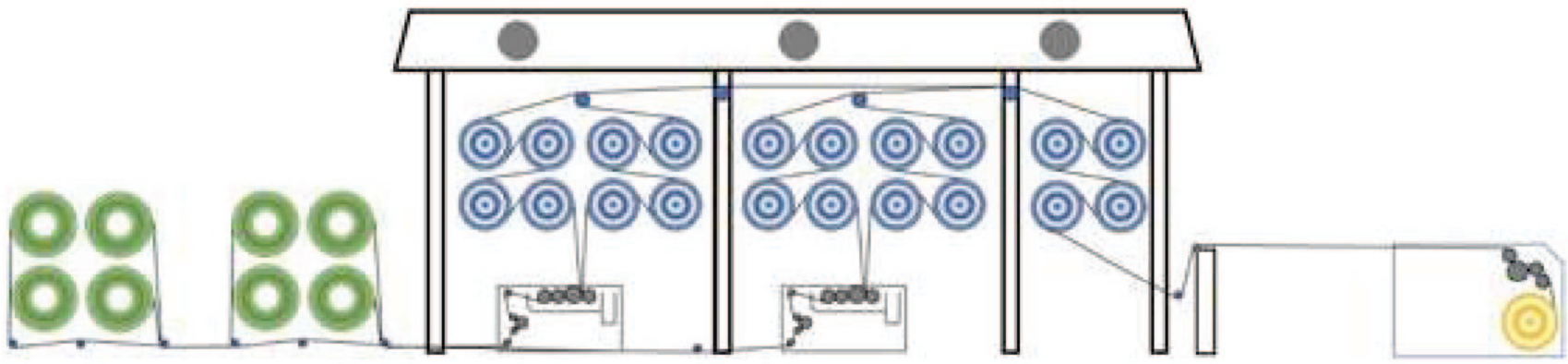
8, equipped with a new fully sealed electric telescopic reed, easy to operate, anti-flying flower;

9. A batch of dividing rod, according to the number of the axis;

10. The ft box adopts gear transmission (company invention patent);

11. Tension auxiliary control system (company invention patent) is used in the dry yarn separation area.

12, with automatic tide return control and one-key speed increase function.



JC739-SF9 型伺服控制系统九单元双浆槽 20 烘筒工艺流程图

JC739-SF9 Process flow chart of double sizing-boxs 20 cylinders sizing machine of servo control system

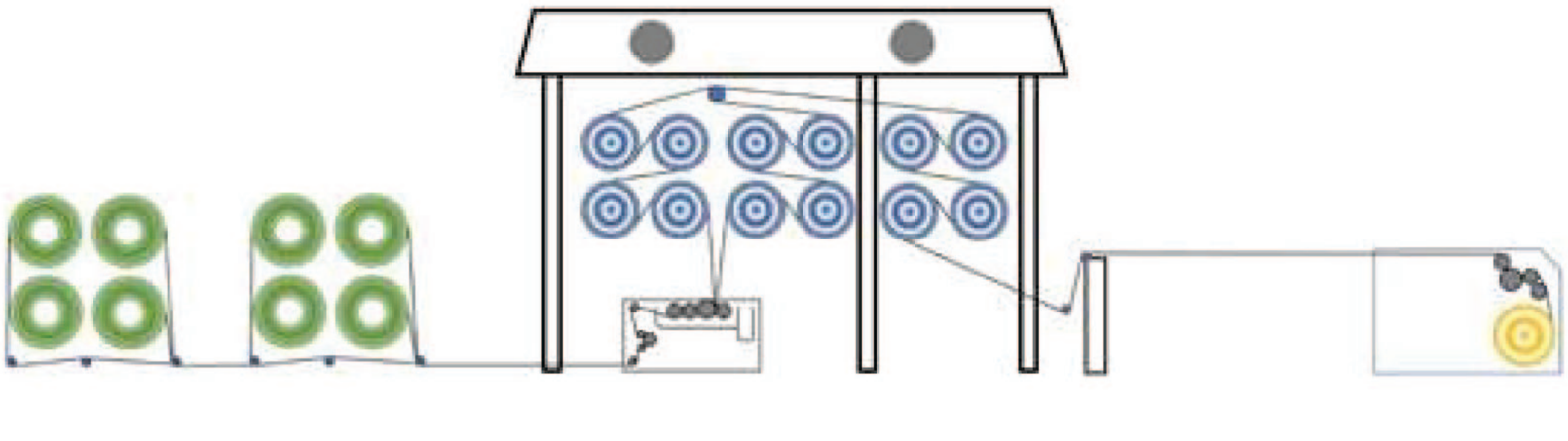
JC739SF-D6 伺服控制系统六单元 传动高速、高效、智能型单浆槽浆纱机

JC739SF-D6 Servo control system six-unit drive high-speed high efficiency intelligent one sizing-box machine

产品的主要技术参数
Main technical parameters of the product

项目 project	单位 Units	HJF739SF-D6
车头有效幅宽 The front is effective and wide	mm	2000/2400/ 2800/3600/4200
适应品种范围 Adapt to variety range		6S-200S 短纤维经纱 Short fiber through yarn
车速 speed	m/min	1 — 200
蜗牛速度 Snail speed	m/min	0.5
织轴卷径 Max weaer’ beam diameter	mm	800/1000/1250
最大卷绕张力 Max. around tension	KN	7.5
最大压力 max. pressure	KN	55
经轴幅宽 Warper’sbeam diameter	mm	1800/2000/2400/2800
烘筒数量 The number of drying cylinder		12/14
容纳经轴数量 Number of warper’s beams		16
经轴最大直径 Max. diameter through warper’s beams	mm	800/1000/1250
重量（约） Weight (approx)	T	25





JC739SF-D6

伺服控制系统六单元单浆槽浆纱机 12 烘筒工艺流程图

Process flow chart of six unit one sizing-box 12 cylinder of servo control system

JC739-S9

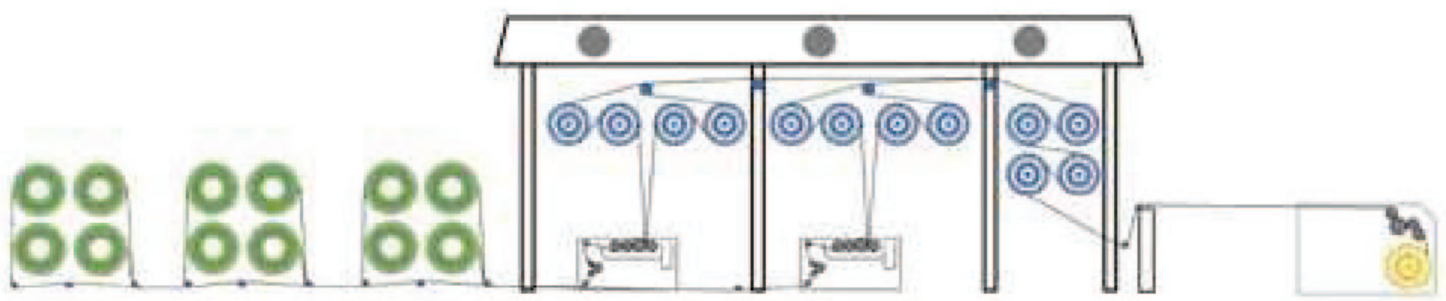
型变频控制系统
九单元机电仪一体化双浆槽浆纱机

JC739-S9 Inverter control system nine-unit electromechanical integrated double sizing-boxs sizing machine

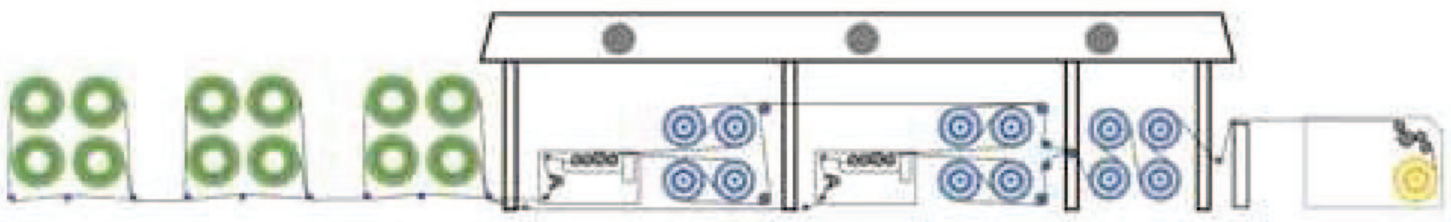
产品的主要技术参数
Main technical parameters of the product

项目 project	单位 Units	JC739-SF9
车头有效幅宽 The front is effective and wide	mm	2000/2400 2800/3600/4200
适应品种范围 Adapt to variety range		6S-120S 短纤维经纱 Short fiber through yarn
车速 speed	m/min	1 — 130
蜗牛速度 Snail speed	m/min	1
织轴卷径 Max weaer’ beam diameter	mm	800/1000/1250
最大卷绕张力 Max. around tension	KN	7
最大压力 max. pressure	KN	55
经轴幅宽 Warper’sbeam diameter	mm	1800/2000/2400/2800
烘筒数量 The number of drying cylinder		14
容纳经轴数量 Number of warper’ s beams		24
经轴最大直径 Max. diameter through warper's beams	mm	800/1000/1250
重量（约） Weight (approx)	T	35





JC739-S9 型双浆槽 12 烘筒高架烘房浆纱机工艺流程图
Process flow chart of the elevated drying room sizing machine of double sinzing-boxs 12 cylinder



JC739-S9 型双浆槽 12 烘筒平行烘房浆纱机工艺流程图
Process flow chart of parallel drying room sizing machine of double sizing-boxs 12 cylinder

JC739-D6

型变频控制系统
六单元单浆槽浆纱机产品技术配置

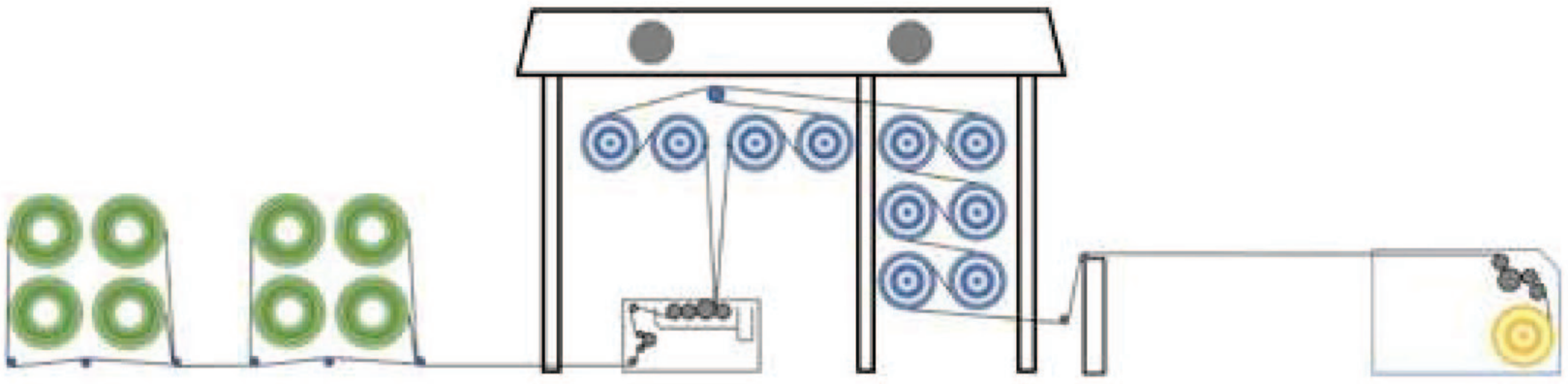
JC739-D6 Six-units of the frequency conversion control system technical configuration of ones sizing-box size machine

产品的主要技术参数
Main technical parameters of the product

项目 project	单位 Units	HJF739-D6
车头有效幅宽 The front is effective and wide	mm	2000/2400 2800/3600
适应品种范围 Adapt to variety range		6S-120S 短纤维经纱 Short fiber through yarn
车速 speed	m/min	1 — 130
蜗牛速度 Snail speed	m/min	1
织轴卷径 Max weaer’ beam diameter	mm	800/1000/1250
最大卷绕张力 Max. around tension	KN	7
最大压力 max. pressure	KN	55
经轴幅宽 Warper’sbeam diameter	mm	1800/2000/2400
烘筒数量 The number of drying cylinder		10
容纳经轴数量 Number of warper’s beams		16
经轴最大直径 Max. diameter through warper’s beams	mm	800/1000/1250
重量（约） Weight (approx)	T	25







JC739-D6

高架烘房 10 烘筒单浆槽浆纱机工艺流程图

Process flow chart of one sizing-box 10 cylinder sizing machine

FAV Group of Companies

+92 305 3097543 | +92 300 2246155

Head Office
B-66/A, S.I.T.E., Area,
Karachi-75700, Pakistan.
+92 (21) 32550093-98
info@vawda.com

Lahore Office
307/E, Nadir street, Ferry Meadowlane,
West canal Bank, Lahore-54590, Pakistan.
+92 (42) 36113239
lhr@vawda.com

Faisalabad Office
P-68, Vawda House, Vawda Street 4,
Zia Town Faisalabad-38000, Pakistan.
+92 (41) 8524576
fsd@vawda.com

Multan Office
Plot # 62/A, Phase-1 Industrial
Estate Multan-60000, Pakistan.
+92 (61) 6538131
mul@vawda.com